Applications of logarithmic functions

1 Search engines give every web page on the internet a score (called a Page Rank) which is a rough measure of popularity/importance. One such search engine uses a logarithmic scale so that the Page Rank is given by: $R = \log_{11} x$, where x is the number of views in the last week.

a Determine the Page Rank of a web page that received 7300 views in the last week. Round your answer to the nearest integer.

b Google uses a base-10 logarithmic scale to get a web page's Page Rank: $R = \log_{10} x$. How many more times the views did a web page with a Page Rank of 5 get than one with a Page Rank of 3?

2 The Palermo impact hazard scale is used to rate the potential for collision of an object near Earth. The hazard rating P is given by the equation $P = \log R$, where R represents the relative risk of collision. Two asteroids are identified as having a relative risk of collision of $\frac{6}{7}$ and $\frac{4}{5}$ respectively. Find the exact difference in their measure on the Palermo impact scale.

3 As elevation A (in metres) increases, atmospheric air pressure P (in pascals) decreases according to the equation: $A = 15\,200\,(5 - \log P)$.
Trekkers are attempting to reach the $8850\ \text{m}$ elevation of Mt. Everest's summit. When they set up camp at night, their barometer shows a reading of $45\,611\ \text{pascals}$. How many more vertical metres do they need to ascend to reach the summit? Round your answer to the nearest metre.

4 The Richter scale is used to measure the intensity of earthquakes. The formula for the Richter scale rating of a quake is given by: $R = \log x - \log a$, where a is the intensity of a minimal quake that can barely be detected, and x is a multiple of the minimal quake's intensity.
The given table shows how quakes are categorised according to their Richter scale rating:

	Richter rating
Minor	$2 - 3.9$
Light	$4 - 4.9$
Moderate	$5 - 5.9$
Strong	$6 - 6.9$

a Rewrite the formula as a single lograithm.

b A seismograph measures the intensity of a quake to be $x = 5711a$.

i Find the Richter scale rating R of this quake to one decimal place.

ii In which category does the quake fall?

c A seismograph measures the intensity of an earthquake to be $15\,850$ times the intensity of a minimal quake.

i Find the Richter scale rating R of this quake to one decimal place.

ii In which category does the quake fall?

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5 The Richter Scale is a base 10 logarithmic scale used to measure the magnitude of an earthquake, given by $R = \log_{10} s$, where s is the relative strength of the quake. Some past earthquakes and their Richter scales are shown in the table below:

Earthquake	Richter scale
Sumatra 2004	9.1
China 2008	7.9
Haiti 2010	7.0
Italy 2009	6.3

a How many times stronger is n earthquake that registers 4.0 on the Richter scale than an earthquake that measures 3.0?

b How many times stronger is a quake of 7.6 than one of 5.2? Round your answer to the nearest whole number.

c How many times stronger was the earthquake in Sumatra compared to the earthquake in China? Round your answer to the nearest whole number.

d The aftershock of an earthquake measured 6.7 on the Richter Scale, and the main quake was 4 times stronger. Find the magnitude of the main quake on the Richter Scale, to one decimal place.

6 The decibel scale, which is used to record the loudness of sound, is a logarithmic scale.

- In the decibel scale, the lowest audible sound is assigned the value of 0.
- A sound that is 10 times louder than 0 is assigned a decibel value of 10.
- A sound 100 (10^2) times louder than 0 is assigned a decibel value of 20.

In general, an increase of 10 decibels corresponds to an increase in magnitude of 10. The table shows the decibel value for various types of noise:

Type of noise	Decibel value
Jet plane	150
Pneumatic drill	120
Industrial	110
Stereo music	100
Inside a car	90
Office	70
Houehold	60
Wind turbine	50
Bedroom	30
Falling leaves	20

a How many times louder is the sound of industrial noise than the sound of a wind turbine?

b If the sound of a normal speaking voice is 50 decibels, and the sound in a bus terminal is 80 decibels, then how many times louder is the bus terminal compared to the speaking voice?

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7 The sound level or loudness, L, of a noise is measured in decibels (dB.), and is given by the formula: $L = 10 \log\left(\frac{I}{A}\right)$, where I (in $\mathrm{watts/cm^2}$) is the intensity of a particular noise and A is the the intensity of background noise that can barely be heard.

a At a concert, standing near a speaker exposes you to noise that has intensity of about $I = 0.5 \times 10^{13} A$.

i How many decibels is this? Round your answer to the nearest dB.

ii Noises measuring up to $85\ \mathrm{dB}$ are harmless without ear protection. By how many decibels does the noise at a concert exceed this safe limit? Round your answer to the nearest dB.

b The maximum intensity which the human ear can handle is about $120\ \mathrm{dB}$. The noise in a recycling factory reaches $132.9\ \mathrm{dB}$. How many times louder than the maximum intensity is the factory noise? Round your answer to one decimal place.

c If one person talks at a sound level of $60\ \mathrm{dB}$, find the value of L which represents the decibel level of 100 people, each talking at the same intensity as that one person.

d If a sound intensity doubles, by how much does the level of sound in decibels increase?

e Given $A = 10^{-16}\ \mathrm{watts/cm^2}$, find:

i The sound level of a sound with intensity $I = 10^{-5}\ \mathrm{watts/cm^2}$.

ii The sound intensity of a passenger plane passing over houses prior to landing, if the engine's loudness is registered at $103\ \mathrm{dB}$. Give an exact answer.

8 The time taken (t years) for A grams of a radioactive substance to decompose down to y grams is given by: $t = -\frac{1}{k} \log_{2.3}\left(\frac{y}{A}\right)$, where k is a constant related to the properties of a particular substance.
Consider the following substances, giving your answers to the nearest year:

- R_1 has a constant of $k = 0.00043$
- R_2 has a constant of $k = 0.00047$

a Find how long it takes 140 units of R_1 to decompose down to 105 units.

b Find how long it takes 200 units of R_2 to decompose down to 150 units.

c Find the half-life of each substance. That is, determine how long it takes a quantity of a substance to decompose down to half the original quantity.

d Will it take twice the half-life for each substance to decompose completely? Explain your answer.

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9 pH is a measure of how acidic or alkaline a substance is. The pH (p) of a substance can be found according to the formula: $p = -\log_{10} h$, where h is the substance's hydrogen ion concentration.
The pH scale goes from 0 to 14, with 0 being most acidic, 14 being most alkaline and pure water has a neutral pH of 7.

a Store-bought apple juice has a hydrogen ion concentration of about $h = 0.0002$. Find the pH of the apple juice correct to one decimal place.

b Is the apple juice acidic or alkaline?

c A banana has a pH of about 8.3. Find h, its hydrogen ion concentration. Give your answer as an exact value.

10 The information entropy H (in bits) of a randomly generated password consisting of L characters is given by: $H = L \log_2 N$ where N is the number of possible symbols for a character in the password.
A case sensitive password consisting of seven characters is to be made using letters from the alphabet and/or numerical digits.

a Find the value of L.

b Find the value of N.

c Find the value of H correct to two decimal places.

d It was found that a seven character password resulted in an entropy of 28 bits. Find the possible number of symbols for a character.

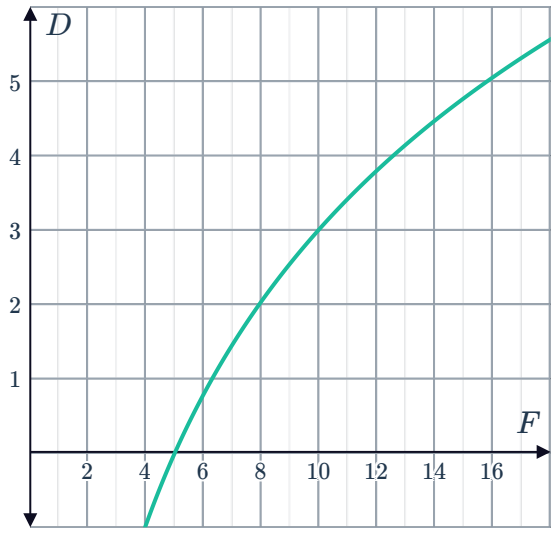
11 The signal ratio D (in decibels) of an electronic system is given by the formula: $D = 10 \log\left(\frac{F}{I}\right)$, where F and I are the output and input powers of the system respectively.

a Find the input power I in megawatts if the output power is equal to 10 MW and the signal ratio is 20 decibels.

b The given graph shows the equation $D = 10 \log\left(\frac{F}{5}\right)$, where the input power is 5 MW. State the interval that contains:

i The signal ratio when $F = 9$.

ii The output power when the signal ratio is $D = 1$.



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12 Consider the function $y = 8\,(2)^{2x}$, for $x \geq 0$.

a The function above can be written as $\log_2 y = mx + k$. Solve for the values of m and k.

b Graph $\log_2 y$ against x.

c Find the rate of change of the linear function.

d Solve for x when $y = 16$.

Applications of exponential functions

13 An initial chemical reaction results in a chain of chemical reactions that follow. The total number of reactions after t seconds is given by the equation $y = 4^t$.

a How many chemical reactions have occurred after the 3rd second?

b How many chemical reactions occur in the 7th second only?

c How many chemical reactions occur in the 8th second only?

d How would you describe the rate of increase of the number of reactions?

14 Consider the table of values:

Number of days passed (x)	1	2	3	4	5
Population of shrimp (y)	5	25	125	625	3125

a Is the number of shrimp increasing by the same amount each day?

b Find the equation linking population, y, and time, x, in the form $y = a^x$.

c Sketch a graph of the population over time.

15 In a laboratory, the number of bacteria in a petri dish is recorded, and the bacteria are found to triple each hour.

a Complete the table of values:

Number of hours passed (x)	0	1	2	3	4
Number of bacteria (y)	1				

b Find the equation linking the number of bacteria (y) and the number of hours passed (x).

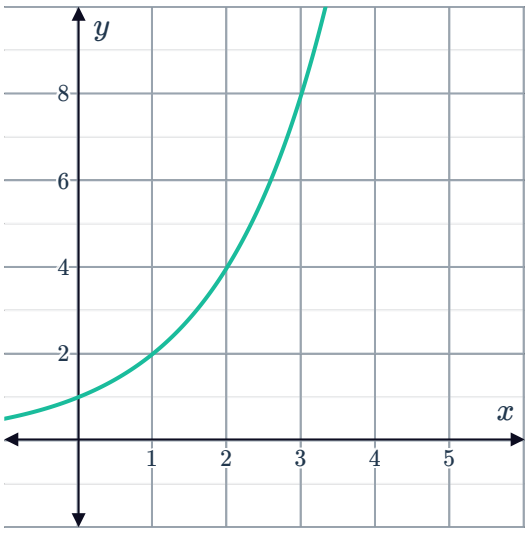
c At this rate, how many bacteria will be present in the petri dish after 11 hours?

d Sketch the graph of the number of bacteria over time.

e Interpret the meaning of the y-intercept in context.

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16 Determine if each the relationships described below could be represented by the given graph:



a The number of people (y) attending a parent/teacher conference when there are x parents, each bringing 2 children.

b The number of layers (y) resulting from a rectangular piece of paper being folded in half x times.

c The number of handshakes (y) made by x people in a room if every person shakes hands with every other person.

17 The number of bacteria in a colony doubles every second.

a Complete the table of values for the first 5 seconds:

Seconds passed (t)	0	1	2	3	4	5
Number of bacteria (y)	1					

b Find an equation for the number of bacteria, y, after t seconds.

c Find the number of bacteria in the colony after 1 minute. Express your answer in exponential form.

d The number of bacteria needs to be graphed against the number of hours h that have passed. Find an equation for the number of bacteria, y, after h hours.

18 1 gram of sugar is poured into a cup of water and immediately begins to dissolve. Each second the amount of sugar remaining is $\frac{1}{4}$ of the amount present in the previous second.

a Complete the table of values:

Seconds passed (t)	0	1	2	3	4	5
Undissolved sugar in grams (y)						

b Find an equation linking undissolved sugar, y, and time, t, in the form:

i $y = a^t$ **ii** $y = a^{-t}$

c Find the difference in undissolved sugar between the first and second seconds.

d Find the difference in undissolved sugar between the second and third seconds.

e Describe the change in the amount of undissolved sugar over time.

f According to this model, will all the sugar eventually dissolve? Explain your answer.

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19 A particular radioactive isotope decays at a rate such that each month there is $\frac{1}{5}$ as much of the isotope as the previous month.

a Given there is a 1 kg of the isotope to start with, complete the table below showing how much of the isotope is left after each month:

Month (x)	0	1	2	3
Amount remaining in kilograms (y)	1			

b Write the model for the amount y of the isotope is remaining after x months.

c How much will be remaining after 7 months?

d Sketch the graph of the amount of isotope over time.

e Interpret the meaning of the y-intercept in context.

20 The magnitude of an earthquake is measured by values on the Richter scale. A 1 unit increase on the Richter scale represents a 10-fold increase in the strength of the earthquake, so that a 4.0 earthquake is 10 times larger than a 3.0 earthquake. The function relating the Richter scale measure x and microns of ground motion y is given by: $y = 10^x$.

a Complete the table of values:

Richter scale measure (x)	1	2	3	4	5
Microns of ground motion (y)	10				

b Graph the function.

c According to the model, can the microns of ground motion ever be 0? Explain your answer.

d Two earthquakes are measured on the Richter scale, one measuring 1 and another measuring 3. How many times larger was the earthquake measuring 3?

e Chile experienced an earthquake followed by an aftershock. The earthquake itself measured 4 and the aftershock measured 3.8 on the Richter scale. How many times stronger was the initial earthquake? Round your answer to one decimal place.

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21 One person in a city is infected with a virus. During the first day, before they start to show symptoms and decide to stay home sick, they infect five more people with the virus.

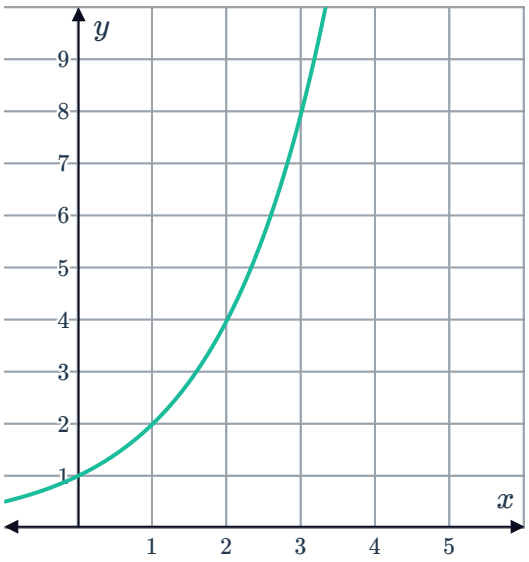
a Given each new person infects an average of five new people on their first day of being sick, complete the table below showing the number of newly infected people for that day:

Day	0	1	2	3	4
Number of people infected	1				

b If this trend continues, write an expression for the number of people infected on day n.

c There are $345\,800$ people in the city in total. Given the trend continues, after how many days will the entire city have been infected?

22 The number of layers (y) resulting from a rectangular piece of paper being folded in half x times is shown in the graph:



a Find the equation linking, y and x, in the form $y = a^x$.

b Interpret the meaning of the y-intercept in context.

c If a rectangular piece of paper is folded 10 times, find the resulting number of layers.

d If a rectangular piece of paper of thickness 0.02 mm is folded 11 times, find the total resulting thickness.

23 The prize of a scratch ticket gives the winner two options:

- Option 1: Receive one payment of $\$500\,000$ immediately, or
- Option 2: Receive $\$1$ on the first day of winnings, double that ($\$2$) on the second day, and so on for three weeks.

a For the second option find the day on which the amount received will first exceed the value of the first option.

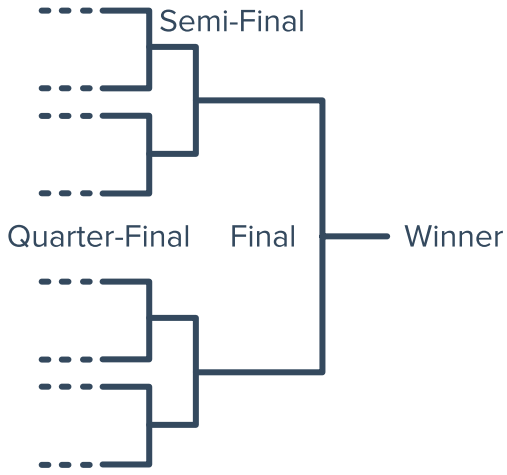
b How much more money does the winner receive if they take the second option?

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24 In a knockout squash tournament, the winner of each round progresses to the next round until there are only two players left. The diagram shows the draw for the final, semi-final and quarter final rounds:

a Complete the table of values for the total number of players, p, that the competition can accommodate given a number of rounds r:

Number of rounds (r)	3	4	5	6
Number of players (p)	8			



b Organisers of a squash tournament want to make sure that each round of the tournament has every spot filled. For what values of p can a tournament be formed?

c Find the equation relating r and p.

d The organisers of a tournament can fit in 10 rounds of play. How many players can they accept into the tournament?

25 When a coin is flipped once there are two possible outcomes: heads (H) or tails (T). And if it is flipped twice there are four possible outcomes: HH, HT, TH or TT.

a How many possible outcomes are there if the coin is flipped three times?

b How many possible outcomes are there if the coin is flipped 8 times?

c Write an expression that represent the number of outcomes when a coin is flipped n times.

26 The population, A, of aphids in a field of potato plants t weeks after intial observation is modelled by: $A = 1000 \times 2^t$.

a State the initial aphid population.

b Find the predicted aphid population after 5 weeks.

c Find the predicted aphid population 2 weeks before initial observations.

27 The growth of a population of mice modelled by $P = 20\,(3^x)$, where P is the population after x weeks. After how many weeks, x, will the population of mice have grown to 4860?

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28 The number of fungal cells, N, in a colony after t hours is given by: $N = 5000\,(4^t)$.

- **a** State the initial population of fungal cells.
- **b** Find the population of fungal cells after 8 hours.
- **c** Sketch the graph of fungal cell population over time.
- **d** How many hours will it take for the population to reach 3 times the original population? Give your answer to two decimal places.

29 The mass in kilograms, M, of a baby monkey at n months of age is given by the equation: $M = 1.8 \times 1.3^n$, for ages up to $n = 6$ months.

- **a** Find the mass, M, of a baby monkey at 5 months of age. Give your answer to one decimal place.
- **b** By how much has the monkey's weight increased between 5 and 6 months of age? Give your answer to one decimal place.

30 The value, A, of a $\$29\,000$ investment at 11% p.a. compounded annually, after n years, can be modelled by: $A = 29\,000 \times 1.11^n$.

- **a** Calculate the value of the investment after 2 years.
- **b** Complete the table:

n	1	2	3	4	5	6
A						

- **c** Describe the rate of change of the investment.

31 A fixed-rate investment generates a return of 6% per annum, compounded annually. The value of the investment is modelled by $A = P\,(1.06)^t$, where P is the original investment. Find the value of the investment after after $3\frac{1}{4}$ years if the original investment was $\$200$.

32 The value, A, of a term deposit in dollars after n years is given by: $A = 9000\,((1.075)^n)$.

- **a** State the initial value of the term deposit.
- **b** By what percentage is the investment increasing by each year?
- **c** Find the value of the term deposit after 6 years to the nearest dollar.
- **d** Sketch the graph of the value of the term deposit over time.
- **e** How many years will it take for the value to reach 4 times the original value? Give your answer to the nearest year.

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33 Maria purchased an artwork for $\$2000$ as an investment. At the end of each year its value increases by 7% of its value at the beginning of the year.

a Find an equation to model the value, V, of the artwork t years after purchase.

b Find the expected value 5 years after purchase to the nearest dollar.

34 In a laboratory, a drug is tested on a sample of 1728 cancer cells. The number of cancer cells is recorded and the cancer cells are found to halve each week.

a Complete the table of values:

Number of weeks passed (x)	0	1	2	3	4
Number of cells (y)	1728				

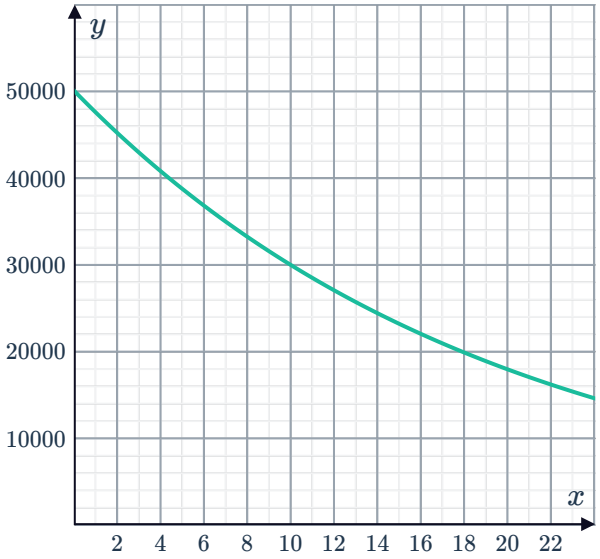
b The equation relating x and y is of the form $y = ba^{-x}$. State the values of a and b.

c According to this model, does the drug completely remove all the cancer cells? Explain your answer.

35 Describe the relationship between an amount N and time t, for each of the following:

a $N = 100\,(0.98)^t$ **b** $N = 200\,(1.07)^t$ **c** $N = 50\,(1.25)^t$ **d** $N = 700\,(0.87)^t$

36 Peter received a lump sum payment of $\$50\,000$ for an insurance claim. Every month, he uses 5% of the remaining funds. The funds, y, after x months is shown in the graph:



a How much would he use in the first month?

b How much would be left after the first month?

c From the graph, estimate the amount left after one and a half years?

d From the graph, estimate for the number of months until the amount reaches $\$30\,000$?

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37 A laptop battery loses 3% efficiency each month.

- **a** Find the percentage of the original battery efficiency remaining after 2 months to two decimal places.
- **b** Find the percentage of the original battery efficiency remaining after 9 months to two decimal places.
- **c** Find the equation that relates the percentage of battery efficiency, E, with the number of months passed, t.
- **d** Sketch the graph of percentage battery efficiency over 30 months.
- **e** From the graph, estimate the number of months it will take for the battery to reach 50% of its original efficiency.

38 A new appliance is valued at $\$990$. Each year it is worth 9% less than the previous year's value.

- **a** Find the value of the appliance after the first year.
- **b** Find the value of the appliance after the second year.
- **c** Find the equation that relates the value of the appliance, A, with the number of years passed, t.
- **d** Find the value of the appliance after twelve years.

39 A radioactive isotope decays by 12% after 50 days. Given an intial amount of $120\ \text{mg}$ of the isotope, find:

- **a** The mass of the isotope after 50 days to two decimal places.
- **b** The mass of the isotope after 100 days to two decimal places.
- **c** The equation that relates the remaining mass of the isotope, m, with the number of days passed, t.
- **d** The remaining mass of the isotope after 120 days to two decimal places.
- **e** Sketch the graph of remaining mass of the isotope over time.
- **f** What is the half-life (the number of days until the mass is half of the original mass) of the isotope? Give your answer to the nearest day.

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40 Consider the population of a country given in the table below. The relationship between the population in millions (P) and the number of years after 1998 (n) can be approximately modelled by: $P = ka^n$.

Year	Population (millions)
1998	57
2003	75
2008	98
2013	127
2018	161

a Use the population in 1998 to estimate the value of k.

b Use the population in the year 2008 to estimate the value of a to two decimal places.

c Using the answers to part (a) and (b) write the equation expressing P in terms of n.

d Complete the table below, using the model in part (c) to estimate the population. Round all answers to two decimal places.

Year	Population (millions)	Estimated population (millions)	Estimated population - actual population (millions)
1998	57		
2003	75		
2008	98		
2013	127		
2018	161		

e Use the model in part (c) to predict the population (in millions) in the year 2058 to two decimal places.

f Do you think the prediction given in part (e) is likely to be accurate? Explain your answer.

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41 A goat farm has 2 breeding goats and 9 non-breeding goats at the start of the first year. The population of breeding goats doubles every year.
The total population of goats can be modelled by the equation $P = a^x + k$, where x is the number of years past.

a Complete the table of values:

Year	1	2	3	4
Number of breeding goats	2			16
Number of non-breeding goats	9	9	9	
Total number of goats	11			

b Find the value of k.

c Find the value of a.

d Assuming none of the goats die, how many goats will there by after 14 years?

e Given the farm begins selling goats when the population exceeds 700, in which year will they start selling goats?

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